

Continuous Improvement and Waste Analysis for Flyscreen Production using Lean Thinking

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Introduction

The company currently produces one standard fly-screen size, and the system is built around a single, predictable product. This works for simple flow, but it limits the ability to handle change.

The company will introduce a second rectangular frame size and offer two mesh materials. As a result, the system will face more changeovers, irregular flow, higher WIP, greater scheduling complexity and longer lead times. These create mura, muri and muda across the value stream [1], making the current system unsuitable for mixed-model production without a Lean redesign.

This report analyses the current-state value stream map, identifies wastes and bottlenecks and shows why the present system cannot handle variation. A future-state Lean design is then proposed using SMED, Kanban, heijunka, standard work and visual management to reduce lead times, improve flow and support mixed-model production as part of ongoing improvement.

Current-State Value Stream Analysis

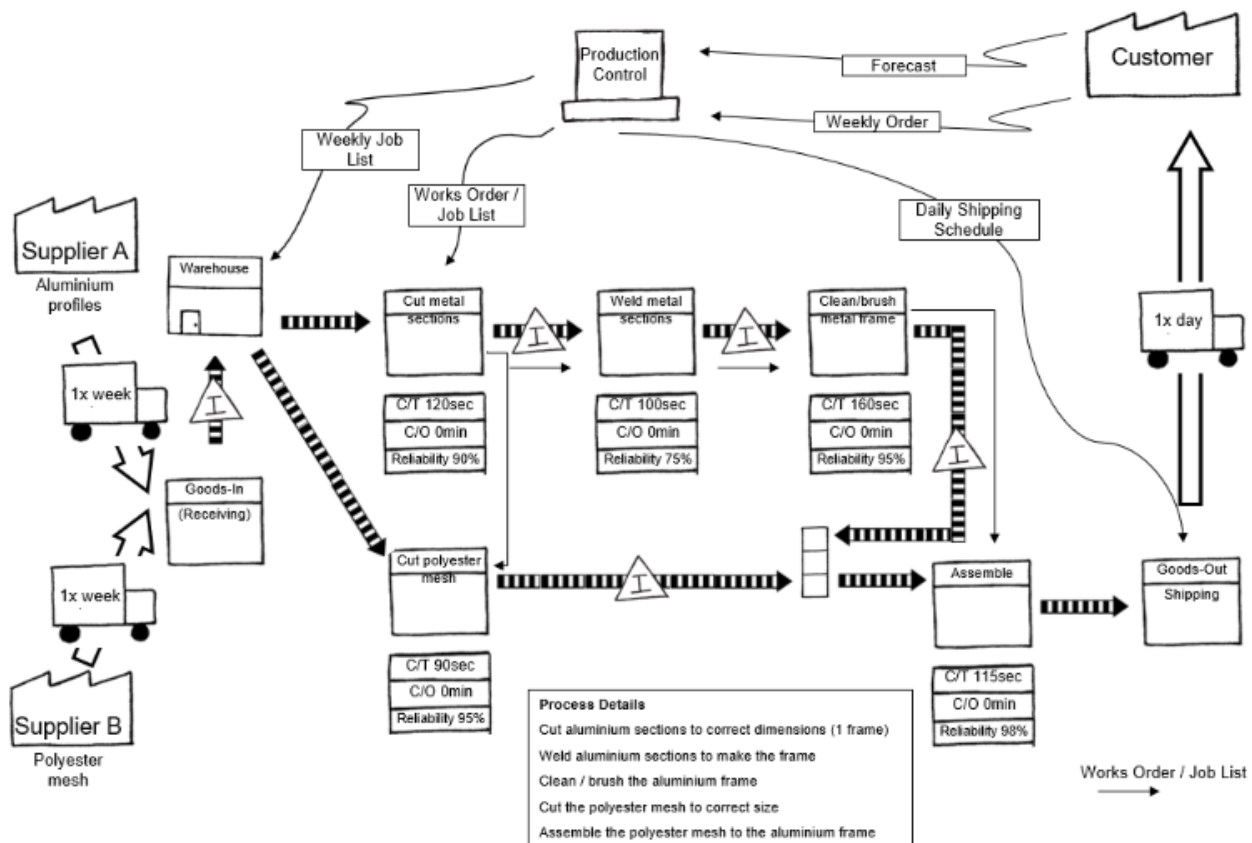


Figure 1: Value Stream Map (current state)

The current system has five steps: cut metal, weld frame, clean frame, cut mesh and assemble. Cleaning has the longest cycle time and is the bottleneck. Welding has low reliability, which interrupts flow and increases waiting. Other steps are more stable, but flow is uneven because WIP builds between stages. Each step works as an isolated island with different cycle times, causing queues, especially before cleaning. The two material flows meet late in the process, which makes coordination harder. With no pull system, work is pushed in batches, creating long lead times and unstable output. This works for a single product, but any variation will increase disruption.

The value stream shows clear evidence of waste (muda, also known as miida), unevenness (mura) and overburden (muri) [1]. Large piles of WIP indicate long lead times and mask defects. Overproduction occurs because the weekly schedule pushes work regardless of downstream readiness. Unevenness comes from low welding reliability and mismatched cycle times. Cleaning creates overburden because it is slower than the other steps.

Information flow also has weaknesses. Production Control issues a weekly schedule that pushes work into the system rather than responding to real demand. There is no pull mechanism, no Kanban and no supermarkets, so each step produces according to the plan instead of consumption. Customer demand arrives daily, but production only changes weekly, causing mismatches and unnecessary WIP. Because there is no pacemaker, the flow is not coordinated. These issues create long lead times, unpredictable flow and poor responsiveness.

Overall, the current value stream is limited by the cleaning bottleneck, welding unreliability, high WIP and batch-driven scheduling. The two material flows join late, causing inconsistent arrival at assembly. The system will not support mixed-model production without a redesigned Lean flow.

Impact of New Product Variants

Introducing new variants increases size and material variation, which adds changeovers to metal cutting, mesh cutting and assembly. These interruptions increase downtime and unevenness. As different variants enter the system, WIP will increase because arrival times and flow rates become less predictable.

Variation also increases scheduling difficulty. A weekly plan cannot adjust to daily changes in demand for each variant, which will create imbalances and more WIP. Matching the correct frame and mesh becomes harder with four combinations, raising the risk of waiting and assembly errors. The cleaning bottleneck will also receive a more irregular mix of work. These issues show that the current system cannot support mixed-model production without a more stable Lean design [2].

Future-State Lean Design

The future state focuses on instability, bottleneck pressure and batch flow, aiming to create smoother movement of parts, reduce WIP and give each process a more stable workload.

The future state must improve flow by addressing key barriers within the present process. Cleaning is the slowest step and receives an unstable stream of parts due to welding reliability, so it needs protection. The flow must be arranged, so the bottleneck receives a controlled and steady supply of work, which will reduce waiting time and stop upstream stages from overwhelming it and causing muri. Welding reliability is also causing the flow to break down early, and the future state will aim to stabilise this step, so it no longer disrupts work. Uneven cycle times create queues and gaps, and for mixed-model production to work the process times need to be closer together. The frame and mesh flows also meet late at assembly, which can cause delays if they arrive at different times. In the future state, both flows need to move at a more synchronised pace. This can be supported through controlled release of work and supermarkets, which will improve flow stability. Cycle times will be brought closer together by improving the method at the bottleneck and reallocating small tasks to welding, which currently has a shorter cycle time. This increases the cycle time of welding and reduces the cycle time of cleaning, bringing these steps closer together and making the flow more even. The future state also reduces unnecessary motion by simplifying the movement of materials and ensuring operators do not travel between separate paths to collect parts, which further supports smoother flow.

The current system relies on pushed work, but the future state will introduce a pull system supported by supermarkets and a pacemaker. Flow will be based on actual consumption rather than a weekly plan, which will give the company more flexibility and protection against changes in demand. Firstly, a supermarket will be placed before the cleaning stage; this protects the bottleneck, limits WIP, prevents overloading and creates a stable feed. Additionally, a supermarket will be placed before the final assembly stage; this ensures assembly never stops, never gets the wrong parts and ensures upstream stages do not overproduce [2]. The supermarkets will use a Kanban signal to keep them stocked at the correct level. Assembly will be assigned as the pacemaker because it is the process closest to the customer and receives the demand information [2]. The pacemaker will support mixed-model production by setting the rhythm for the entire production system. All upstream processes will respond to what assembly requires, with parts being pulled towards it, this will stabilise the flow. Heijunka will level the release of work at the pacemaker, by mixing the different frame and mesh variants in small regular batches [3]. This will smooth the demand seen by each process and prevent sudden peaks and troughs in the workload, and a more even release of work makes the pull

system more predictable. Overall, this redesign will replace the weekly push plan with a controlled pull system that reduces overproduction, controls WIP and provides a stable flow suited to mixed-model production.

With the introduction of mixed-model production, changeovers will be introduced to both the metal and mesh cutting stages. Metal cutting will have to stop to switch between frame sizes, and mesh cutting will have to stop to switch between materials. There will also be a changeover at the assembly stage to allow for switching between the four frame and mesh combinations. To minimise disruption to flow and support more frequent switching between variants, the future state will apply SMED principles to reduce setup times [4]. Internal and external activities will be separated; internal work is only possible once the machine is stopped, while external work can be carried out while the machine is running. The aim of separating these activities is to shift as much work as possible to external tasks and minimise downtime. Standardised tools, fixtures and settings will be used, which allows operators to switch between variants with fewer adjustments. By reducing the time and complexity involved, SMED will allow smaller batches, smoother flow and more frequent switching between variants, which is essential for mixed-model production to work.

The redesign must address quality issues and improve process stability, so the system can support variation. Currently, welding has a 75% success rate, meaning its output is very inconsistent. This unreliability creates unevenness and unpredictable arrival times and creates waste in the form of scrap and defects. The future state must find a way to stabilise this stage; therefore, a root-cause analysis must be carried out to identify what is causing the welders issues and how they can be helped. Additionally, standard work will be implemented [5], meaning operators will be using clear, simple and repeatable steps, this will reduce variation between workers and make root causes easier to detect. Mixed-model production also increases the risk of assembly errors, as operators must match the correct frame and mesh combination. Poka-Yoke will support mixed-model production by preventing operators from selecting the wrong frame or mesh during assembly. Simple error-proofing devices and clear OK/No Good checks will stop incorrect combinations from entering the flow, which reduces defects, this will improve reliability and help maintain stable flow as variation increases [6]. Visual management will help operators see abnormalities and priorities immediately. Colour coding and clear labelling will show the correct frame and mesh combinations and reduce mistakes, support standard work and improve reliability [7]. The future state will use Jidoka principles to build quality into the process [8], with in-process checks, ensuring problems are detected early and are not hidden in WIP. When defects are detected, the system will stop, allowing the operator to correct the issue and prevent it from recurring. These changes create stable and predictable processes, which are vital for the integration of mixed-model production.

The future state is not intended to be perfect; it does represent a step forward from the current system. It forms part of Lean's PDCA cycle [5] and provides a starting point for ongoing continuous improvement rather than a one-off redesign.

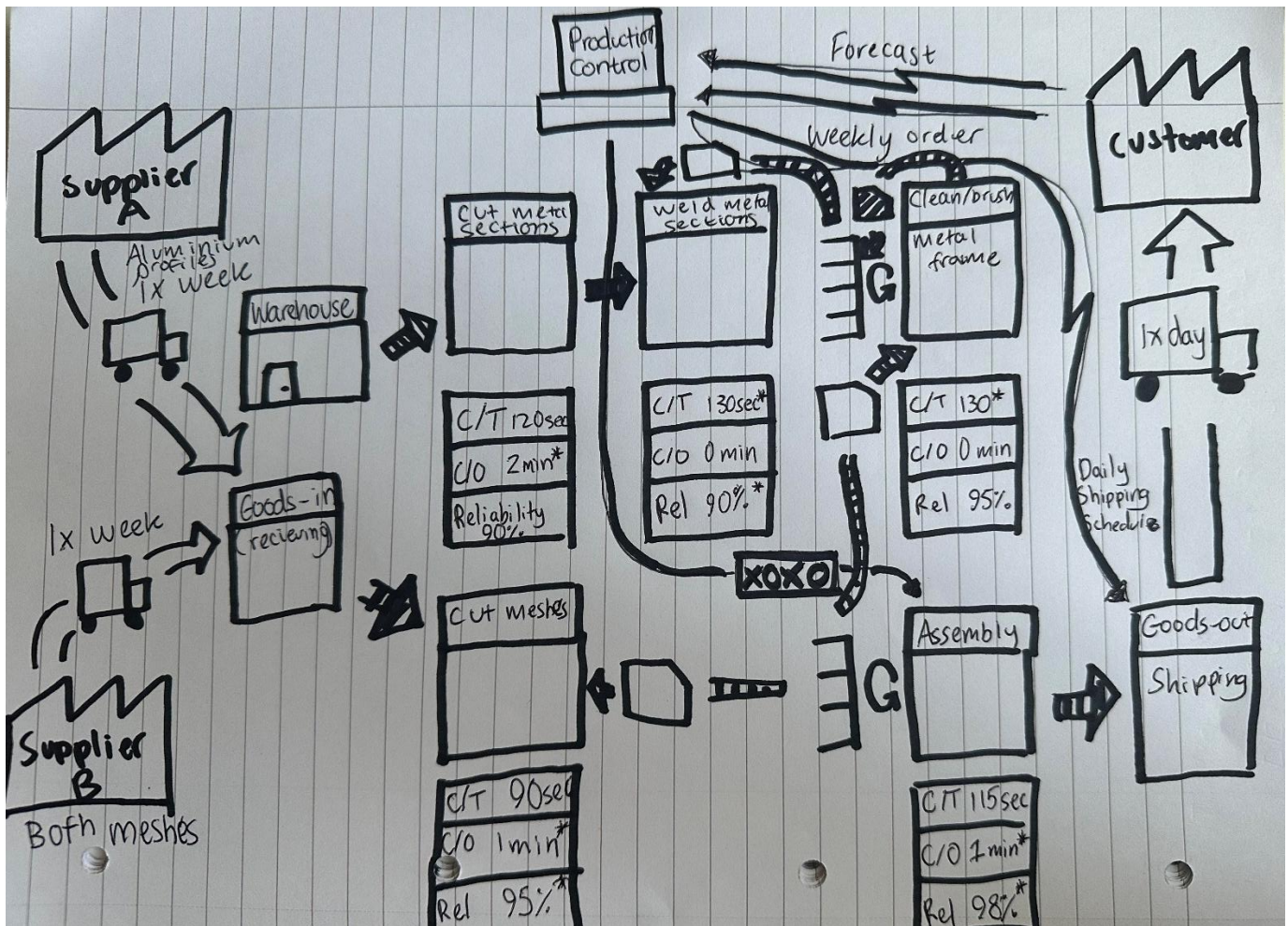


Figure 2: Future State Value Stream Map, note *: estimation

Critical Discussion

The proposed future state focuses on the main issues shown in the current-state map and uses Lean tools to remove their root causes. Cycle times will be brought closer together through simple line balancing, which reduces the unevenness seen in the present system. The supermarket before cleaning controls WIP and provides a steady flow of inputs to the bottleneck, preventing the instability caused by batch releases. Assembly will act as the pacemaker, replacing the weekly push schedule and creating a predictable pull signal for the rest of the process. A levelled release of variants will reduce peaks and dips in workload and help maintain steady flow as mixed-model production is introduced. SMED will support this by shortening changeovers and allowing smaller batches without disrupting flow. Jidoka, standard work and Poka-Yoke will address the increased risk of errors by making problems easier to detect and prevent. Together, these changes reduce waste, increase stability and make the system better suited for mixed-model production.

The future state is a step forward rather than a complete solution and will need close monitoring as variation is introduced. Its success relies on operators following standard work and understanding the Kanban pull signals, as any deviation will affect flow and stability. Welding reliability remains one of the biggest risks, especially as small additional tasks may increase complexity, although the new quality controls should help counter this. The supermarkets reduce WIP between steps, but they can still hide issues for short periods, which means defects must be detected early through Jidoka and Poka-Yoke. The design represents the next stage in the PDCA cycle and should be refined as the system collects more performance data, as there is a risk that management may view these changes as a one-off fix rather than the start of continuous improvement.

Conclusion

The current-state analysis shows unstable flow, high WIP between stations, unreliability and a bottleneck at cleaning. The introduction of variants will amplify these issues if they are left unchanged. The future state addresses these problems by aligning cycle times and creating a more level flow. A pull system, supported by supermarkets and a pacemaker, will control WIP and provide predictable movement of work. Quality controls will be built into the process to reduce errors and improve reliability. This redesign provides a foundation for further improvement as the system develops.

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